

JEE MAIN REVISION MASTER: MOVING CHARGES AND MAGNETISM 2026 4 QUESTIONS 6.67% WEIGHTAGE ↓19.93%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 48 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Biot-Savart Law & Ampere's Circuital Law	28 PYQs	Magnetic Field
Very High	Lorentz Force, Cyclotron & Magnetic Force on Wires	21 PYQs	Magnetic Forces

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Biot-Savart Law & Ampere's Circuital Law	$dB = \frac{\mu_0 I dl \sin\theta}{4\pi r^2}, \quad B_{\text{wire}} = \frac{\mu_0 I}{2\pi d}, \quad B_{\text{center}} = \frac{\mu_0 N I}{2R}, \quad B_{\text{solenoid}} = \mu_0 n I$
Lorentz Force, Cyclotron & Magnetic Force on Wires	$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}), \quad r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}, \quad T = \frac{2\pi m}{qB}, \quad \frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Biot-Savart Law & Ampere's Circuital Law (Very High Yield — 28 PYQs)

Core Principle: Calculation of magnetic fields generated by steady currents in wires, loops, and solenoids.

■ **Caution / JEE Pitfall:** In solenoid formula $B = \mu_0 n I$, n is number of turns per UNIT LENGTH ($n = N/L$), not total turns N .

■ **Speed Trick:** Right-hand thumb rule: thumb points along current, curled fingers give direction of circular magnetic field lines.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Two identical circular current loops carrying equal currents are placed with their axes inclined at 45° to each other as shown in t..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "A solenoid is 1 m long and 4 cm in diameter. It has five layers of windings of 1000 turns each and carries a current of 7A. The magnetic fie..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Identify the correct statement..." [Ans: N/A]

3.2 Lorentz Force, Cyclotron & Magnetic Force on Wires (Very High Yield — 21 PYQs)

Core Principle: Trajectory of charged particles in magnetic fields, velocity selector ($\vec{E} \perp \vec{B}$), and force between parallel currents.

■ **Caution / JEE Pitfall:** Magnetic force $\vec{F} = q(\vec{v} \times \vec{B})$ does ZERO work because $\vec{F} \perp \vec{v}$. It changes direction of velocity, never kinetic energy!

■ **Speed Trick:** Parallel currents in the SAME direction ATTRACT; anti-parallel currents REPEL.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A point charge is placed in a moving train. A passenger A sitting in the train and person B on the ground observe the fields due to this cha..." [Ans: N/A]
- **[2026 Shift] Q#3 (MCQ):** "Biot-Savart law indicates that an electron moving with a velocity $\vec{v}$ produces a magnetic field $\vec{B}$ around it such that..." [Ans: N/A]
- **[2026 Shift] Q#4 (MCQ):** "A proton, an electron and an α -particle enter at right angles to a uniform magnetic field with the same velocity. If R_p, R_e a..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A point charge is placed in a moving train. A passenger A sitting in the train and person B on the ground observe the fields due to this charge. Then

- (A) observes both electric and magnetic fields
- (B) observes both electric and magnetic fields
- (C) observes only magnetic field
- (D) observes only electric field

Correct Answer: None

Solution Analysis: The correct option is: Option B: B observes both electric and magnetic fields The point charge is placed in the moving train. For passenger A sitting inside the train, the charge is at rest. A stationary charge produces only an electric field around it. So A observes only electric field, not magnetic field. For person B standing on the ground, the train is moving. So the charge is also moving with the train. A moving charge is equivalent to an electric current, and current produces a magnetic field. Since the charge itself also has electric field, B observes both electric and magnetic

Benchmark Problem 2: KCET 2026 (Q#2)

Two identical circular current loops carrying equal currents are placed with their axes inclined at 45° to each other as shown in the figure. The resultant magnetic field at P is

- (A) $\frac{\mu_0 I}{6\sqrt{2}} R [(\sqrt{2}+1)\mathbf{i} + \mathbf{j}]$
- (B) $\frac{\mu_0 I}{6\sqrt{2}} R [\sqrt{2}\mathbf{i} + \mathbf{j}]$
- (C) $\frac{\mu_0 I}{6R} [(\sqrt{2}+1)\mathbf{i} + \mathbf{j}]$
- (D) $\frac{\mu_0 I}{6R} [\sqrt{2}\mathbf{i} + \mathbf{j}]$

Correct Answer: None

Solution Analysis: The magnetic field at a point on the axis of a circular current loop is $B = \frac{\mu_0 I R^2}{(R^2 + x^2)^{3/2}}$. Here, R is the radius of the loop and x is the distance of point P from the centre of the loop along its axis. For both identical loops, point P is at the same distance from the centre. From the figure, this distance is $x = \sqrt{3}R$. So, the magnetic field due to each loop has the same magnitude: $B_1 = B_2 = \frac{\mu_0 I R^2}{(R^2 + (\sqrt{3}R)^2)^{3/2}}$
 $B_1 = B_2 = \frac{\mu_0 I R^2}{(R^2 + 3R^2)^{3/2}}$ $B_1 = B_2 = \frac{\mu_0 I R^2}{(4R^2)^{3/2}}$